ITD2104 – Practical 1: Mini Smart-Room Agent

Student: Maftuna Sharabbayeva

Part A – Analyse a real intelligent system: Gmail spam filter

Question	Answer
1. Name of the system	Gmail spam filter
2. Type and why	ML system, mainly supervised learning: it learns from labelled examples of spam and non-spam emails.
3. Goal - what does it try to achieve?	Classify incoming email and identify messages that should be treated as spam.
4. Layer 1 – What data does it collect?	Email text, sender information, message metadata and user feedback.
5. Layer 2 – how is the data cleaned / what features?	Clean and transform email data into features suitable for classification; handle noisy or irrelevant information.
6. Layer 3 – what rules or models does it use?	A trained machine-learning classification model; systems may also use rules.
7. Layer 4 – what decision does it take?	Predict spam/not-spam and decide whether the message should be filtered.
8. Layer 5 – how does the decision reach the user or the world?	Move or label the message in the mailbox, including the spam folder.
9. Feedback - how could it learn from results?	Use user actions such as marking a message spam/not spam to improve future classification.

Part C – Data drift experiment

Run	Energy	Wasted	Violations	Faulty removed
1) Baseline	48.0 kWh	0.0 kWh	1	8
2) Drift: sensor reads +3 °C	31.8 kWh	0.0 kWh	18	8
3) Drift: class moved to 18:00	49.5 kWh	3.0 kWh	2	8
4) Bonus: learned occupancy	49.5 kWh	0.0 kWh	0	8

1. Run 2: The +3 °C sensor drift makes the agent believe the room is warmer than it really is. It therefore heats less, reducing energy use but increasing comfort violations.

2. Run 3: The real afternoon class moves to 18:00 while the agent still expects 14:00. Heating is wasted in the empty room around the old class time, while the 18:00 class starts too cold.

3. Automatic detection: Monitor sensor residuals/calibration against trusted readings for sensor drift; compare observed occupancy with timetable predictions to detect schedule/occupancy drift.

Improvement idea

Keep the five-layer design but add adaptive occupancy learning and sensor-drift monitoring. The bonus model uses the five previous days and treats occupancy probability > 0.5 as expected use.

